LECTURE 28: MATH1061/7861
TOPICS: ELEMENTARY PROPERTIES OF GROUPS
DATE: TUESDAY MAY 5, 2026

Learning objectives:

(1) Gain more familiarity with groups and examples of groups.

(2) Learn the definition of a subgroup of a groups.

A group means we have a set G together with a binary operation $* : G \times G \to G$. satisfying associativity, unit, inverse.

aka identity

A subset $H \subseteq G$ is a subgroup, we have to check ① closure

② unit

③ inverses.

identity is 1 in this case

Q1: Which of the following is a subgroup of $(\mathbb{Z}_7 - \{[0]\}, \times)$? Justify your answer.

(a) $\{1, 2\}$

$[2]\cdot[2] = [4] \notin \{[1], [2]\}$

Not closed So not a subgroup.

(b) $\{1, 2, 4\}$

Is a subgroup

$4 \cdot 4 = 16 = 2$

$2 \cdot 2 = 4$

$2 \cdot 4 = 4 \cdot 2 = 8 = 1.$

closure holds.

$2^{-1} = 4$, $4^{-1} = 2$ inverse holds

(c) $\{1, 4\}$

$4 \cdot 4 = 16 = 2$ Not closed.

Not a subgroup.

Yes.

(a)

Q2: Is $(\mathbb{Z}, +, 0)$ a subgroup of $(\mathbb{Q}, +, 0)$? Justify your answer.

identity: $0 \in \mathbb{Z}$ ✓

closure: sum of two integers is an integer ✓

inverse of x is $-x$. if x is an integer so is $-x$. ✓

(b) Let X be the subset of $\mathbb{Q}$ consisting of all rational numbers in the form $\frac{a}{b}$ where b is a power of 2. $a, b \in \mathbb{Z}, b \neq 0$ which can be written

Is $(X, +)$ a subgroup of $\mathbb{Q}$? Yes.

✓ identity: $0 = \frac{0}{2} = \frac{0}{4} = \cdots$

✓ closure: $\frac{a}{2^x} + \frac{b}{2^y} = \frac{a \cdot 2^y + b \cdot 2^x}{2^{x+y}} \in X$

✓ inverse: $-\frac{a}{2^x} = \frac{-a}{2^x} \in X$.

$\mathbb{Z} \subseteq X \subseteq \mathbb{Q}$

Subgroups

(3) Give examples of a group G together with a subgroup H.

(1) $(\mathbb{Q}, +, 0) \subseteq (\mathbb{R}, +, 0)$

subgroup.

(2) $(\mathbb{Q} - \{0\}, \times, 1) \subseteq (\mathbb{R}^{\times} - \{0\}, \times, 1)$

(3) Is $(\mathbb{Z}_6, +)$ a subgroup of $(\mathbb{Z}_7, +)$

$\mathbb{Z}_6 = \{[0], [1], [2], [3], [4], [5]\}$

all integers divisible by 6

integers ~~divisible~~ have remainder 1 modulo 6

$\mathbb{Z}_7 = \{[0], [1], [2], [3], [4], [5], [6]\}$

all integers divisible by 7

$\mathbb{Z}_6$ is not a subset of $\mathbb{Z}_7$
(So it's not a subgroup either)